📘 Exception Handling in Java

1️⃣ What is Exception Handling?

Exception Handling in Java is a mechanism to handle runtime errors so that the program does not crash and normal flow is maintained.

- Example Errors:

Division by zero
Invalid array index
User-defined condition (like age < 18)

2️⃣ Types of Exceptions

Type	Meaning	Example
Checked Exception	Checked at compile time	IOException
Unchecked Exception	Occurs at runtime	ArithmeticException

Program uses:

Custom Exception (AgeSmall)

ArithmeticException (divide by zero)

3️⃣ try Block

- Meaning:

Contains risky code that may cause an exception.

- Syntax:

```
try {
    // risky code
}
```

- In Code:

```
try {
    int age = 16;
    Check.checkAge(age);
    int result = a / b;
```

```
}
```

If any error occurs → control jumps to catch block.

4 catch Block

🔹 Meaning:

Handles the exception and prevents program crash.

🔹 Syntax:

```
catch(ExceptionType e) {
    // handling code
}
```

🔹 Multiple Catch Syntax:

```
try {
    // risky code
}
catch(Type1 e) {
}
catch(Type2 e) {
}
```

🔹 Ins Code:

```
catch (AgeSmall e) {
    System.out.println("Custom Exception: " +
e.getMessage());
}

catch (ArithmeticException e) {
    System.out.println("Cannot divide by zero!");
}
```

5 finally Block

🔹 Meaning:

Always executes whether exception occurs or not.

🔹 Syntax:

```
finally {
    // always executed code
}
```

🔹 In Code:

```
finally {
```

```
    System.out.println("Finally block always executes.");
}
```

✓ Runs in both cases (exception or no exception)

6 throw Keyword

🔹 Meaning:

Used to manually create and throw an exception.

🔹 Syntax:

```
throw new ExceptionType("Error message");
```

🔹 In Code:

```
if (age < 18) {
    throw new AgeSmall("Age must be 18 or above to vote.");
}
```

This manually generates the exception.

7 throws Keyword

🔹 Meaning:

Declares that a method may throw an exception.

🔹 Syntax:

```
returnType methodName() throws ExceptionType {
    // code
}
```

🔹 In Code:

```
static void checkAge(int age) throws AgeSmall
```

This means:

➡️ This method may generate AgeSmall exception.

8 User-Defined Exception

🔹 Meaning:

Creating your own custom exception class.

🔹 Syntax:

```
class CustomException extends Exception {
    CustomException(String msg) {
        super(msg);
    }
}
```

- In Code:

```
class AgeSmall extends Exception {
    AgeSmall(String msg) {
        super(msg);
    }
}
```